

Graph Cross-Domain Continual Fine-Tuning via Orthogonal LoRA Routing with Contrastive Expert Specialization

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Abstract

This paper investigates a novel and critical problem of *Graph Cross-Domain Continual Fine-Tuning*, which aims to adapt a large pre-trained Graph Foundation Model across diverse domains. Existing continual graph learning methods are mostly limited to incremental settings within only a single domain, and are typically trained from scratch. As a result, they fail to handle cross-domain shifts effectively, suffer from severe forgetting, and lack transferability. To address these challenges, we present **G-CORMoL**, Graph Continual Fine-tuning with **Orthogonal, Router-driven Mixture of LoRA** experts. G-CORMoL achieves effective adaptation while preserving prior knowledge by enforcing mathematical orthogonality between expert LoRA adapters, thereby eliminating interference across tasks. It further supports cross-domain knowledge transfer through a symmetric dual-driven routing mechanism that learns a global composition policy over all learned LoRA experts. In addition, it promotes expert specialization via a contrastive objective with theoretical guarantees. Extensive experiments on different cross-domain task orders demonstrate that G-CORMoL achieves robust state-of-the-art performance, not only preventing catastrophic forgetting but also leveraging accumulated knowledge to enable positive transfer.

CCS Concepts

• **Information systems** → **Data mining**; • **Computing methodologies** → *Transfer learning*.

Keywords

Graph Machine Learning, Transfer Learning, Continual Learning

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1 Introduction

The machine learning paradigm has shifted from training models *from scratch* to a prevailing *pre-train, then fine-tune* methodology. This trend, driven by the success of Large Language Models (LLMs) [4, 14, 15, 35], is now reshaping graph machine learning with the emergence of large-scale Graph Foundation Models (GFM) [38, 40, 46]. Pre-trained on vast multi-domain graph datasets, these models serve as strong backbones that capture generalizable structural knowledge. As a result, adapting such large pretrained models to downstream tasks has become the prevailing mainstream for achieving state-of-the-art performance.

However, this paradigm faces a central challenge: the dynamic nature of real-world graph data [11, 24]. Applications rarely involve a single, static graph domain. Instead, models are expected to adapt to a continuous stream of evolving graph domains. This exposes an important **Research Gap**: most traditional continual graph learning studies have focused only on incremental changes within a single domain. Typical examples include learning new node classes or related tasks in the same domain [19, 25, 42, 44, 47]. Moreover, most of these studies are still developed under a training-from-scratch paradigm, which fails to exploit the generalizable knowledge embedded in GFMs and is inadequate for addressing significant cross-domain shifts.

The limitations of these settings, together with the requirements of real-world applications, underscore the importance of exploring a new direction for GFMs. We therefore formalize a necessary and novel research problem: **Graph Cross-Domain Continual Fine-Tuning**. As illustrated in Figure 1 (a), consider a large pretrained GFM deployed in practice. It may first be adapted to academic citation networks (e.g., Cora), then to co-purchase networks (e.g.,

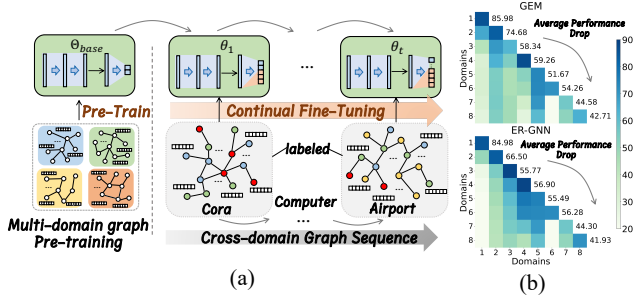


Figure 1: (a) Illustration of the Graph Cross-Domain Continual Fine-Tuning setting. (b) Performance of the traditional continual learning method (GEM) and the state-of-the-art graph continual learning method (ERGNN) when evaluated on cross-domain datasets in a continual fine-tuning setting.

Computer), and later to transportation networks (e.g., Airport). Each domain exhibits distinct structural properties, ranging from homophily to sparsity, yet they also share transferable patterns such as community organization or hierarchical structures. In such scenarios, both bare fine-tuning and existing state-of-the-art continual graph learning methods usually suffer from sharp performance drops and severe forgetting when evaluated in cross-domain settings, as shown in Figure 1 (b). This motivates a critical research question: *How can a powerful pre-trained GFM be efficiently and effectively adapted across a sequence of diverse graph domains, not only avoiding catastrophic forgetting but also leveraging accumulated knowledge to facilitate positive transfer?*

Addressing this question raises two key challenges: (i) *cross-domain shifts*, where structural shifts across domains make it difficult for a single model to adapt efficiently across a sequence of diverse tasks while retaining accuracy in a parameter-efficient manner; (ii) *knowledge retention and transfer*, where the goal extends beyond merely preventing catastrophic forgetting to actively reusing all learned for positive transfer.

To tackle these challenges, we introduce **G-CORMoL** (Graph Continual Fine-tuning with **O**rthogonal, **R**outer-driven **M**ixture of LoRA Experts). Motivated by the efficiency and modularity of LoRA, G-CORMoL integrates orthogonal subspace construction for interference-free adaptation with a Mixture-of-Experts architecture guided by a symmetric dual-driven router. The router not only learns globally consistent expert composition policies but also promotes functional specialization through contrastive regularization with theoretical guarantees. Together, these mechanisms transform isolated adapters into a collaborative expert system capable of intelligent cross-domain transfer. The main contributions of our paper are as follows:

- We introduce and address the novel problem of *Graph Cross-Domain Continual Fine-Tuning* for large-scale pre-trained GFMs, bridging the gap between GFMs and dynamic multi-domain learning environments.
- We design G-CORMoL, which combines (i) effective and efficient adaptation through *Orthogonal Knowledge Isolation* and *Graph LoRA Adaptation* within frozen subspaces, ensuring provable preservation and parameter efficiency; (ii) *Symmetric Dual-Driven Routing* that integrates utility prediction with contrastive expert

specialization under theoretical guarantees, enabling dynamic and globally consistent cross-domain transfer.

- We conduct extensive experiments on diverse and challenging cross-domain graph sequences, demonstrating that G-CORMoL consistently achieves state-of-the-art performance, with strong resistance to forgetting and clear evidence of positive transfer.

2 Related Work

2.1 Continual Graph Learning

Methods for Continual Graph Learning (CGL) are typically grouped into three main strategies. Replay-based methods store and rehearse a small subset of past data to alleviate forgetting [1, 9, 16, 23, 33, 42, 44, 45, 47]. Regularization-based approaches, such as EWC [13], add a penalty to the loss function to discourage significant changes to important weights from prior tasks [3, 13, 18, 30, 36]. Finally, parameter isolation strategies assign new, dedicated modules to each task and freeze the old ones, which perfectly preserves past knowledge but leads to unbounded model growth [41, 43]. Distinct from these classic paradigms, recent work has shifted towards more parameter-efficient solutions [8, 22]. Some approaches, for instance, have explored prompt-based learning to steer frozen GNNs towards new tasks using small, learnable prompts [22]. Our work extends this efficiency-focused direction. We build on the principle of parameter isolation but leverage Parameter-Efficient Fine-Tuning (PEFT). By creating lightweight, task-specific LoRA experts, we achieve robust knowledge preservation without the significant computational and memory overhead of traditional methods.

2.2 Mixture of LoRA for Continual Learning

Recent work in continual learning has explored Mixture-of-Experts (MoE) frameworks to manage a growing library of task-specific LoRA experts [6, 10, 39]. Inspired by successes in LLMs, these methods typically employ a router to select and combine experts, aiming to balance specialization with knowledge transfer. Recent advances, such as LoRAMoE [6], follow this paradigm by freezing the backbone and introducing multiple LoRA experts, with a router assigning mixture weights to them during inference. This improves adaptability and mitigates forgetting, but the router remains a passive gating mechanism without an explicit design for cross-domain knowledge transfer. Our work offers a newer perspective on this paradigm. Instead of treating the router as a passive selection mechanism, we empower it to actively shape the expert space. By incorporating a contrastive learning objective into the router's training, we more effectively promote the specialization and functional diversity of experts, leading to a more modular library and more synergistic knowledge composition.

3 Methodology

3.1 Preliminary

We address the novel and critical problem of continual fine-tuning on a sequence of cross-domain graphs. We begin with a pre-trained Graph Foundation Model (GFM) based on an existing approach, denoted as $f(\cdot; \Theta_{base})$, whose parameters Θ_{base} remain frozen. The model is presented with a sequence of T downstream tasks, $\mathcal{D} = \{\mathcal{D}_1, \mathcal{D}_2, \dots, \mathcal{D}_T\}$. Each task $\mathcal{D}_t$ presents a node-level prediction

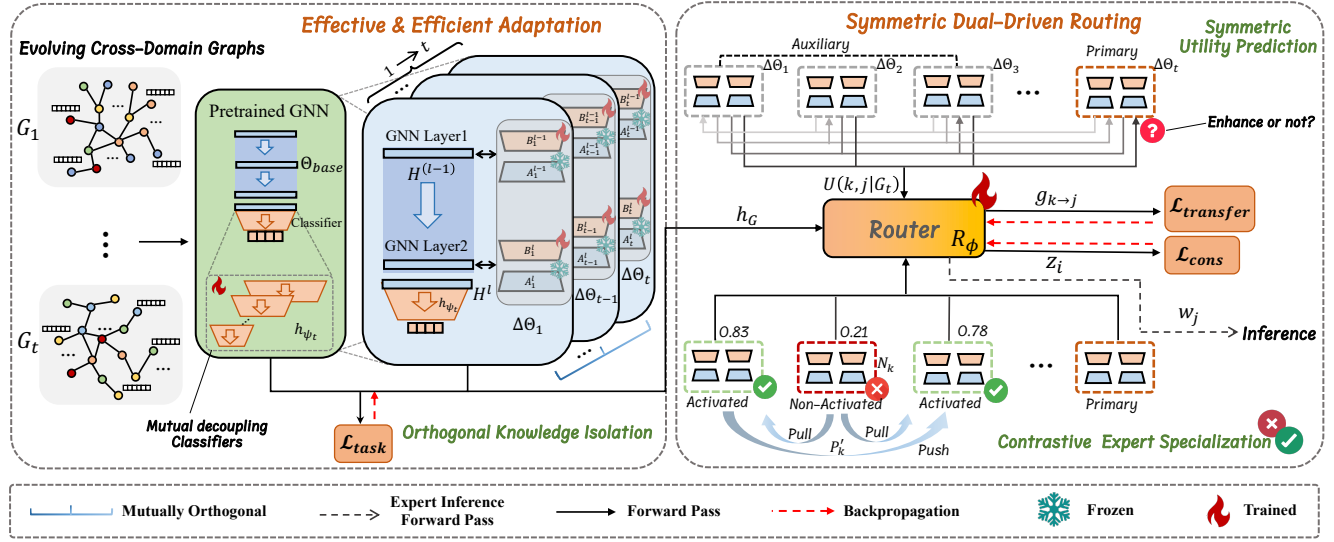


Figure 2: The G-CORMoL framework. For each new task, *Orthogonal Knowledge Isolation* is achieved by training a LoRA expert within an orthogonal subspace, ensuring interference-free preservation of prior knowledge and parameter-efficient learning. A *Symmetric Dual-Driven Router* establishes a global composition policy through *Utility Prediction* and *Contrastive Expert Specialization*, enabling dynamic expert integration at inference to support cross-domain transfer and mitigate forgetting. Each domain retains its own task head in adaptation, and inference on different domains uses their original heads while incorporating the dynamically routing on composed LoRA experts.

problem on a graph $\mathcal{G}_t = (\mathcal{V}_t, \mathcal{E}_t, \mathcal{X}_t)$, where $\mathcal{V}_t$ is the set of N_t nodes, $\mathcal{E}_t$ is the set of edges, and $\mathcal{X}_t \in \mathbb{R}^{N_t \times d}$ is the node feature matrix. The corresponding labels are provided in a vector $\mathcal{Y}_t \in \mathbb{R}^{N_t}$, where each task $\mathcal{D}_t$ has a distinct label set. Accordingly, these tasks arrive sequentially, forming a curriculum designed to introduce diverse and progressively challenging distribution shifts. Consecutive graphs $\mathcal{G}_t$ and $\mathcal{G}_{t-1}$ may differ significantly in their domain, scale, and intrinsic graph properties (e.g., homophily, density), making $P(\mathcal{G}_t) \neq P(\mathcal{G}_{t-1})$, where P is the graph distribution. Upon the arrival of each task $\mathcal{D}_t$, our objective is to adapt the model by learning a small, task-specific set of parameters. Within our framework, besides the specific classification heads decoupled across tasks, this corresponds to a Low-Rank Adaptation (LoRA) module, whose parameters are denoted as $\Delta\theta_t$. For an L -layer GNN, $\Delta\theta_t = \{(A_t^{(l)}, B_t^{(l)})\}_{l=1}^L$, where $A_t^{(l)} \in \mathbb{R}^{d_{l-1} \times r}$ and $B_t^{(l)} \in \mathbb{R}^{r \times d_l}$ are the low-rank decomposition matrices for layer l , with rank $r \ll d_l, d_{l-1}$. After training on task t , the set of all learned adapters is $\mathbb{E}_t = \{\Delta\theta_1, \dots, \Delta\theta_t\}$. The model can leverage θ_{base} in combination with this growing library to make predictions. Therefore, this learning process is guided by two fundamental objectives:

- (1) **Effective & Efficient Adaptation:** The model should effectively adapt to each new task $\mathcal{D}_t$, achieving high predictive accuracy in a parameter-efficient manner, with the overall goal of maximizing the average performance across all domains.
- (2) **Knowledge Transfer:** The model should retain its performance on all previous tasks $\{\mathcal{D}_1, \dots, \mathcal{D}_{t-1}\}$ while leveraging knowledge captured by previous adapters $\{\Delta\theta_1, \dots, \Delta\theta_{t-1}\}$ to improve learning on all learned tasks in inference stage.

To meet these objectives, our framework G-CORMoL consists of two complementary components: (1) *Orthogonal Knowledge Isolation*

combined with *Graph LoRA Adaptation*, which together guarantee interference-free preservation and parameter-efficient learning, thereby enhancing effective and efficient adaptation. (2) *Symmetric Dual-Driven Routing*, which integrates utility prediction with contrastive expert specialization under a symmetric training protocol, enabling dynamic and globally consistent expert composition that drives positive transfer across domains.

3.2 Adaptation with Orthogonal LoRA Experts

3.2.1 Orthogonal Knowledge Isolation. The foundation of G-CORMoL is a structural mechanism that mathematically guarantees preservation of past knowledge by constraining each new LoRA expert to operate within a vector subspace orthogonal to those of earlier experts. This structural separation provides a critical foundation, enabling the subsequent mixture of these orthogonal LoRA experts for effective knowledge transfer.

Let the frozen base GNN have L layers, where the forward pass for layer l is $H^{(l)} = \sigma(H^{(l-1)} W_{base}^{(l)})$. A LoRA expert introduces a task-specific update $\Delta W_t^{(l)} = A_t^{(l)} B_t^{(l)}$, aligned with the design of GraphLoRA [37], and extended in our framework through orthogonal subspace construction to ensure provable isolation. The key to preventing catastrophic forgetting lies in the principled construction of the matrix $A_t^{(l)}$, which defines the expert's update subspace. In our framework, we draw a sharp distinction between the roles of the two LoRA matrices: $A_t^{(l)}$ is a non-trainable matrix that structurally defines the "safe" orthogonal update directions, while only $B_t^{(l)}$ contains trainable parameters that learn the task-specific projection onto this fixed subspace. This design choice is fundamental to our guarantee against interference [34].

Upon the arrival of task $\mathcal{D}_t$, for each layer l , we first define the historical knowledge subspace, $\mathcal{S}_{< t}^{(l)} = \text{span}(\{A_1^{(l)}, \dots, A_{t-1}^{(l)}\})$. Let $\mathbf{S}_{< t}^{(l)}$ be a matrix whose columns form an orthonormal basis for this subspace. Given the input activations $H_t^{(l-1)} \in \mathbb{R}^{N_t \times d_{l-1}}$ for the training nodes, we isolate novel information by projecting these activations onto the orthogonal complement of $\mathcal{S}_{< t}^{(l)}$:

$$\hat{H}_t^{(l-1)} = H_t^{(l-1)} (\mathbf{I} - \mathbf{S}_{< t}^{(l)} (\mathbf{S}_{< t}^{(l)})^T). \quad (1)$$

The projected activations, $\hat{H}_t^{(l-1)}$, contain only the components of the new task's feature space that are linearly independent of all past knowledge subspaces. To find an optimal basis for this novel space, we perform Singular Value Decomposition (SVD) on its covariance matrix to identify the principal directions of variance:

$$U, \Sigma, V^T = \text{SVD}((\hat{H}_t^{(l-1)})^T \hat{H}_t^{(l-1)}). \quad (2)$$

The first r left singular vectors, $U_{:,r}$, form an orthonormal basis for the r -dimensional subspace of greatest variance within the novel information. We set this basis as the new, non-trainable LoRA matrix, $A_t^{(l)} = U_{:,r}$, which is guaranteed by construction to be orthogonal to $\mathcal{S}_{< t}^{(l)}$. The historical knowledge basis is then augmented and re-orthogonalized for the next task via QR decomposition: $\mathbf{S}_t^{(l)} = \text{QR}([\mathbf{S}_{< t}^{(l)}, A_t^{(l)}])$.

By pre-computing and freezing the orthogonal basis $A_t^{(l)}$, our approach offers two key advantages. First, it provides a hard structural guarantee of knowledge isolation, as update directions are constrained by construction rather than being guided by a soft regularization loss, which may not fully prevent interference. Second, it enhances parameter efficiency; by only training the $B_t^{(l)}$ matrix for each new task, we significantly reduce the number of trainable parameters and the computational overhead of adaptation. Moreover, orthogonality is not only essential for isolating past knowledge but also serves as the prerequisite that enables the mixture of LoRA experts to achieve transferable cross-domain knowledge sharing. The necessity of this design is further supported by the ablation study in Table 2, which highlights its critical role in ensuring stable knowledge preservation and reliable cross-domain transfer.

3.2.2 Graph LoRA Adaptation. With the interference-free update subspace for task $\mathcal{D}_t$ established and defined by the frozen matrix $A_t^{(l)}$, the model's adaptation objective is to learn the optimal task-specific projection onto this subspace. This is accomplished by training only the low-rank matrix $B_t^{(l)}$ for each layer l . The model generates predictions by integrating the new expert's update with the base model's output. For each layer l , the forward pass is:

$$\mathbf{H}^{(l)} = \sigma \left(f^{(l)}(\mathcal{G}_t, \mathbf{H}^{(l-1)}; \Theta_{\text{base}}^{(l)}) + \mathbf{H}^{(l-1)} A_t^{(l)} B_t^{(l)} \right), \quad (3)$$

where the term $\mathbf{H}^{(l-1)} A_t^{(l)} B_t^{(l)}$ represents the low-rank adaptation. The final node representations $\mathbf{H}_{\text{final}}$ are then passed to a task-specific classifier h_{ψ_t} . The adaptation is driven by minimizing the cross-entropy task loss:

$$\mathcal{L}_{\text{task}}(\theta_{B_t}, \psi_t) = - \frac{1}{|\mathcal{V}_t^{\text{train}}|} \sum_{v \in \mathcal{V}_t^{\text{train}}} \log (\text{Softmax}(h_{\psi_t}(\mathbf{H}_{\text{final}, v}))_{\mathcal{Y}_t(v)}), \quad (4)$$

where the optimization targets only the new expert's parameters $\theta_{B_t} = \{B_t^{(l)}\}_{l=1}^L$ and the task-specific classifier's parameters ψ_t .

This LoRA-based design achieves efficiency by significantly decreasing the volume of trainable parameters compared with conventional full fine-tuning. It is also effective, since optimization is confined to the orthogonal subspace tailored for each new task $\mathcal{D}_t$, thereby enabling strong task performance without interference. As shown in Figure 5(b), our method achieves this dual objective: maintaining competitive accuracy while requiring significantly less memory, demonstrating both efficiency and effectiveness in our setting. This decoupling of task-specific knowledge into modular LoRA experts forms the foundation for their dynamic composition within a Mixture-of-Experts, thereby enabling targeted and reliable knowledge transfer.

3.3 Symmetric Dual-Driven Routing

To enable effective retention and positive knowledge transfer, we introduce a routing mechanism that dynamically composes the preserved and specialized experts. The router's training is guided by two principles: *symmetry*, to mitigate temporal bias, and *dual-driven optimization*, to learn both what to compose and how to structure the expert space for composition.

Symmetry Principle. A naive router trained solely from the perspective of the current task t would develop a task-biased policy. To counter this, our training protocol is **Symmetric**: during the training phase for task t , we iterate through every available expert $k \in \{1, \dots, t\}$, treating each one in turn as the "primary expert". This encourages the router to learn a global, unbiased composition policy from all possible perspectives, learning not only how to solve the current task but also how historical knowledge can be reapplied. The router, $\mathcal{R}_\phi$, is implemented as a neural network with a sequential three-part architecture: (1) an MLP that processes the graph-level representation $h_{\mathcal{G}} \in \mathbb{R}^{d_{\text{graph}}}$, mapping it into the router's hidden space $h_{\text{graph}} \in \mathbb{R}^{d_{\text{hid}}}$; (2) an expert embedding layer that takes a discrete expert index k as input and outputs a dense vector $e_k \in \mathbb{R}^{d_{\text{hid}}}$, where for a primary-auxiliary pair (k, j) we construct the relational vector $(e_j - e_k)$; and (3) a gate predictor MLP that takes the concatenated vector $[h_{\text{graph}}, e_j - e_k] \in \mathbb{R}^{2d_{\text{hid}}}$ and outputs a scalar passed through a Sigmoid to yield the gate value $g_{k \rightarrow j} \in [0, 1]$. This value reflects the anticipated utility of collaboration, and repeating the process for all $j \neq k$ yields the set of gate values $\{g_{k \rightarrow j}\}$. Together, these components form a unified mechanism: the graph representation provides the task context, the expert embeddings encode expert identities, and the gate predictor integrates them to estimate transfer utility. The transfer loss propagates through the entire router to align predicted utilities with empirical signals, while the contrastive loss primarily regularizes the expert embedding space to encourage specialization. Consequently, once trained, inference is entirely training-free, requiring only a single forward pass to obtain the composition weights.

3.3.1 Learning a Predictor for Collaborative Utility. The first of the router's two training drivers is designed to answer a critical question: *for a given primary expert k , how beneficial is the collaboration of an auxiliary expert j ?* To achieve this, we train the router $\mathcal{R}_\phi$ to act as a utility estimator that learns to predict this utility. The training process unfolds in three steps:

First, for a given graph $\mathcal{G}_t$ and primary expert k , the router predicts a continuous gate value $g_{k \rightarrow j} = \mathcal{R}_\phi(\mathbf{h}_{\mathcal{G}_t}, k)_j$ for each potential auxiliary expert j . Concretely, for each primary–auxiliary pair (k, j) , the router constructs an input vector of dimension $2d_{hid}$ by concatenating the processed graph representation with the relational vector $(e_j - e_k)$ derived from the expert embedding layer, where the subtraction captures their relative difference. This input is then fed into the gate predictor to produce $g_{k \rightarrow j}$, which represents the router’s estimate of the potential benefit.

Second, to provide a supervised signal, we compute the ground truth for this prediction. This is done by calculating the **Empirical Synergistic Utility**, $U(k, j | \mathcal{G}_t)$, which measures the actual performance gain. We explicitly evaluate the model to get the task loss with the primary expert acting alone, $\mathcal{L}_{\text{task}}(f_k)$, and subtract the loss when the auxiliary expert collaborates, $\mathcal{L}_{\text{task}}(f_{k+j})$. The resulting difference, $U(k, j | \mathcal{G}_t) \triangleq \mathcal{L}_{\text{task}}(f_k) - \mathcal{L}_{\text{task}}(f_{k+j})$, is the empirical utility. This pairwise utility assessment represents a principled and computationally efficient approach. The structural orthogonality among the experts provides the justification for this design, as it ensures their individual contributions are largely additive, serving as a reliable proxy for their true collaborative potential.

Finally, the router is trained by minimizing the Mean Squared Error between its prediction $g_{k \rightarrow j}$ and the scaled ground-truth utility. This entire process is averaged over all possible collaborators and, to embody the symmetry principle, is taken in expectation over a uniform choice of primary expert k . The final transfer loss is thus defined as:

$$\mathcal{L}_{\text{transfer}} = \frac{1}{t} \sum_{k=1}^t \mathbb{E}_{\mathcal{G}_t \sim \mathcal{D}_t} \left[\sum_{j \neq k} (g_{k \rightarrow j} - \sigma(U(k, j | \mathcal{G}_t)))^2 \right]. \quad (5)$$

where the sigmoid function $\sigma(\cdot)$ scales the empirical utility to match the router’s $[0, 1]$ output range, and the utility itself is calculated without tracking gradients to create a stable regression target.

3.3.2 Contrastive Regularization for Expert Specialization. While the utility predictor learns to assess the immediate benefit of a collaboration, the second driver of our router has a more structural objective: to actively shape the entire library of experts to become more specialized and functionally distinct over time. This process is achieved through contrastive regularization.

A principled way to formalize this objective is by maximizing a quantity from information theory known as Mutual Information (MI) [31, 32]. In our context, maximizing the MI between an expert’s representation, $\mathbf{z}$, and the graph context, C , encourages the router’s selections to induce a representation space where expert roles are clearly defined. Intuitively, this ensures that knowing an expert’s representation provides significant information about the input graph—a condition that can only be met if the experts are functionally specialized. Directly maximizing MI is intractable. Instead, we use contrastive learning to optimize a tractable lower bound on this objective. The process works as follows:

First, we use the output from the utility predictor to form our training groups. The gate values $\{g_{k \rightarrow j}\}$, which represent confidence scores, are passed through an activation threshold τ . This threshold serves as a reliable supervision signal for grouping experts. Collaborators with a score above the threshold are identified as “positive” partners for the current task, forming the positive set

$\mathcal{P}'_k = \{k\} \cup \{j | g_{k \rightarrow j} > \tau\}$. All other experts become the negative set $\mathcal{N}_k$. A detailed sensitivity analysis of the activation threshold τ is provided in Appendix C.2.

Next, we apply a “pull-push” dynamic on the experts’ underlying representations, denoted as $\mathbf{z}_i \in \mathbb{R}^{d_{hid}}$, where $\mathbf{z}_i$ is the embedding vector of expert i produced by the expert embedding layer of the router. For a given primary expert k , the contrastive loss encourages the representations of all experts within the positive set $\mathcal{P}'_k$ to be pulled closer together in the embedding space. Simultaneously, it pushes their representations away from those of the experts in the negative set $\mathcal{N}_k$. This is achieved using an InfoNCE-style loss function, which for a given expert k is:

$$\mathcal{L}_{\text{cons}}^{(k)} = - \mathbb{E}_{\mathcal{G}_t \sim \mathcal{D}_t} \left[\sum_{i \in \mathcal{P}'_k} \log \frac{\sum_{p \in \mathcal{P}'_k \setminus \{i\}} \exp(\text{sim}(\mathbf{z}_i, \mathbf{z}_p)/T)}{\sum_{a \in (\mathcal{P}'_k \setminus \{i\}) \cup \mathcal{N}_k} \exp(\text{sim}(\mathbf{z}_i, \mathbf{z}_a)/T)} \right], \quad (6)$$

where $\text{sim}(\cdot, \cdot)$ is cosine similarity and T is a temperature hyperparameter. This practical loss is theoretically grounded: minimizing it optimizes a lower bound on the mutual information objective, as stated in Theorem 3.1. The total contrastive loss is the symmetric average $\mathcal{L}_{\text{cons}} = \frac{1}{t} \sum_{k=1}^t \mathcal{L}_{\text{cons}}^{(k)}$, which serves as a structural regularizer on the entire expert library.

THEOREM 3.1. *Let Z be a random variable over expert embeddings $\mathbf{z} \in \mathbb{R}^{d_{hid}}$ produced by the router’s expert embedding layer, and let C be a random variable for the graph context $\mathcal{G}_t$. The mutual information $I(Z; C | \phi)$ is lower-bounded by the objective optimized through $\mathcal{L}_{\text{cons}}$:*

$$I(Z; C | \phi) \geq \mathbb{E} \left[\log \frac{p(\mathbf{z} | \mathcal{G}_t, \phi)}{p(\mathbf{z} | \mathcal{G}_t, \phi) + |\mathcal{N}_k| p(\mathbf{z} | \phi)} \right] - \log(|\mathcal{N}_k| + 1), \quad (7)$$

where the expectation is taken over the data distribution, with positives drawn from $p(\mathbf{z} | \mathcal{G}_t, \phi)$ and negatives from $p(\mathbf{z} | \phi)$, as defined by the router $\mathcal{R}_\phi$. A detailed proof is provided in Appendix A.1.

3.4 Training Objective

Upon the arrival of a new task $\mathcal{D}_t$, the learning process is framed as a unified, constrained optimization problem. Let θ_{B_t} denote the new expert’s trainable LoRA matrices $\{B_t^{(l)}\}_{l=1}^L$. The full set of trainable parameters for this task is $\Omega_{\text{train}} = \{\theta_{B_t}, \phi, \psi_t\}$, where ϕ and ψ_t are the parameters for the router and task-specific classifier, respectively. The training objective combines three distinct components. The first is the primary task loss, $\mathcal{L}_{\text{task}}$, which adapts the new LoRA expert to the current task. The other two components, $\mathcal{L}_{\text{transfer}}$ and $\mathcal{L}_{\text{cons}}$, guide the training of the router. The complete objective is formulated as:

$$\begin{aligned} \min_{\Omega_{\text{train}}} \quad & \mathcal{L}_{\text{task}}(\theta_{B_t}, \psi_t) + \lambda_{\text{transfer}} \mathcal{L}_{\text{transfer}}(\phi) + \lambda_{\text{cons}} \mathcal{L}_{\text{cons}}(\phi), \\ \text{s.t.} \quad & (A_t^{(l)})^T S_{<_t}^{(l)} = 0, \quad \forall l \in \{1, \dots, L\}. \end{aligned} \quad (8)$$

The constraint enforces orthogonality with prior knowledge. The joint objective balances two complementary goals: the task loss adapts the new expert, while the routing losses co-evolve the router and specialize the entire expert library.

3.5 Inference: Dynamic Expert Composition

The inference process directly leverages the router parameters ϕ , which were optimized during the end-to-end symmetric training. Consequently, this stage is entirely training-free. For a given graph $\mathcal{G}_t$ from task t , the system performs a single, training-free forward pass to dynamically compose experts. Since the composition weights have already been optimized during the Symmetric Dual-Driven Routing stage, the inference process simply requires a single forward pass to dynamically integrate the relevant LoRA experts, thereby enhancing positive knowledge transfer across domains. A global graph representation, $\mathbf{h}_{\mathcal{G}_t}$, is extracted and passed to the trained router $\mathcal{R}_\phi$ along with the primary expert index k to obtain utility gates. Based on these, an active set of collaborators $\mathcal{A}_k = \{j \mid g_{k \rightarrow j} > \tau\}$ is formed, and their gates are softmax-normalized into weights $\{w_j\}_{j \in \mathcal{A}_k}$.

Let $\Delta f(\mathcal{G}; \Delta\Theta_j)$ denote the output contribution of a single LoRA module. The final node representation is a composition of the primary expert's full output with a weighted mixture of auxiliary expert residuals:

$$\mathbf{H}_{\text{final}} = \underbrace{f(\mathcal{G}_t; \Theta_{\text{base}})}_{\text{Primary Expert Output}} + \underbrace{\sum_{j \in \mathcal{A}_k} w_j \cdot \Delta f(\mathcal{G}_t; \Delta\Theta_j)}_{\text{Synergistic Auxiliary Composition}} \quad (9)$$

This composed representation $\mathbf{H}_{\text{final}}$ is then passed to the task-specific classification head h_{ψ_t} followed by a softmax to obtain the final class probabilities, where each task retains its own head and inference on past tasks naturally uses their respective heads.

4 Experiments

4.1 Experimental Setup

4.1.1 Datasets. The experimental evaluation consists of two phases: pre-training and continual fine-tuning. The pre-training stage consists of five diverse, large-scale graph datasets: Citeseer [12], DBLP [7], Photo [29], Facebook [26], and GitHub [26]. This selection comprises a variety of graph structures and domains, providing a strong and generalized graph foundation model. For the continual fine-tuning phase, we construct a sequence of eight distinct graph datasets, referred to as Order1: Cora [28], Pubmed [21], CoauthorCS [29], Computer [29], WikiCS [20], LastFMAsia [27], DeezerEurope [27], and Airport [5]. Data order details are provided in Appendix D.2. This evaluation protocol offers a comprehensive way to assess our method's robustness and ability to generalize across diverse domains.

4.1.2 Baselines. We compare G-CORMoL against a comprehensive set of strong baselines in continual learning, which fall into three main categories. For General Continual Learning Methods, we include several foundational strategies that are not specific to graphs: EWC [13], MAS [2], GEM [19], and LWF [17]. For Continual Graph Learning Approaches, we compare against a range of methods designed specifically for graph data: the regularization-based TWP [18]; replay-based methods ER-GNN [47] and SSM [44]; and recent parameter-isolation and prompt-based approaches TPP [22]

and PDGNNs [42]. For Mixture-of-Experts LoRA methods, we include LoRAMoE [6] as a representative PEFT-based baseline. Additionally, we establish performance bounds using two reference models. Fine-Tune serves as a lower bound, where the model is naively updated on each new task, demonstrating catastrophic forgetting. Joint-Training acts as an upper bound, trained with access to all task data simultaneously. Detailed descriptions and configurations for each baseline are provided in the Appendix E.

4.1.3 Implementation Details. To ensure a fair comparison, all methods, including G-CORMoL and all baselines, are initialized from the identical pre-trained GFM. For all baselines and our method, we train each new expert for 200 epochs using the AdamW optimizer with a learning rate of 1×10^{-3} and a weight decay of 5×10^{-4} . For all baseline methods, we strictly adhere to the hyperparameter configurations reported in their original publications to ensure faithful comparisons. The code of G-CORMoL is available at <https://github.com/Qy-ds/G-CORMoL>.

4.1.4 Evaluation Metrics. We evaluate model performance using two primary metrics designed to assess overall effectiveness and to specifically quantify the benefits of our core architectural contributions. **Average Accuracy (AA).** This metric assesses the model's overall performance across all tasks encountered so far. After the model has been trained on the full sequence of T tasks, its accuracy is evaluated on the test set of every task $i \in \{1, \dots, T\}$. Let $A_{T,i}$ denote the accuracy on task i at the end of the entire training sequence. The Average Accuracy is then defined as $AA = \frac{1}{T} \sum_{i=1}^T A_{T,i}$. A higher AA indicates stronger overall performance, signifying both effective adaptation to new tasks and robust retention of knowledge from past tasks. **Average Gain (AG).** To explicitly measure the knowledge transfer ability over time, we further define the Average Gain (AG). This metric quantifies the average improvement on past tasks after the model has learned from the entire sequence, indicating positive backward transfer. Let $A_{k,i}$ be the accuracy on task i after training up to task k . The AG is formally defined as $AG = \frac{1}{T-1} \sum_{i=1}^{T-1} (A_{T,i} - A_{i,i})$. A high positive AG is desirable as it shows that learning new tasks enhances performance on previous ones. For our G-CORMoL framework, AG specifically validates our core contribution: the dynamic expert composition mechanism. It directly quantifies the performance improvement attributable to router-driven expert collaboration.

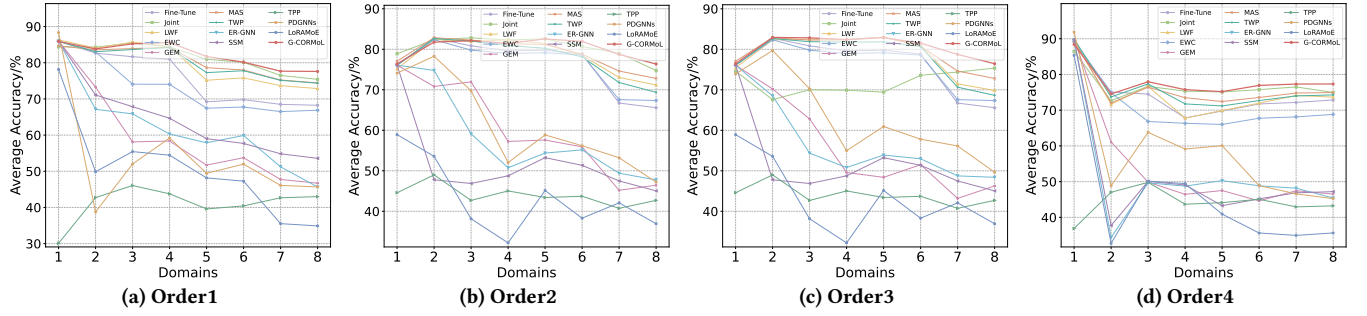
4.2 Experimental Results

In this section, we extensively evaluate our methods on multi-domain sequence datasets to answer the following research questions and demonstrate a fine-grained phenomenon analysis:

- **RQ1:** How does G-CORMoL perform against state-of-the-art baselines in terms of overall accuracy?
- **RQ2:** To what extent does G-CORMoL facilitate positive knowledge transfer, and how does this contrast with baselines that suffer from catastrophic forgetting?
- **RQ3:** What is the contribution of each core component of G-CORMoL: the orthogonal subspaces, the utility predictor, and the contrastive regularizer, to the final performance?
- **RQ4:** How sensitive is the performance of G-CORMoL to the choice of LoRA rank r ?

Table 1: Continual Fine-Tuning performance across different task orders. Bold represents the best-performing results. AA/% $\uparrow$ and AG/% $\uparrow$ indicate higher values are better.

Method	Order 1		Order 2		Order 3		Order 4	
	AA/% $\uparrow$	AG/% $\uparrow$	AA/% $\uparrow$	AG/% $\uparrow$	AA/% $\uparrow$	AG/% $\uparrow$	AA/% $\uparrow$	AG/% $\uparrow$
Fine-Tune	68.82 $\pm$ 0.38	-10.71 $\pm$ 0.40	65.49 $\pm$ 0.12	-15.36 $\pm$ 0.17	69.48 $\pm$ 0.41	-11.45 $\pm$ 0.52	72.89 $\pm$ 0.32	-6.56 $\pm$ 0.17
LWF	73.90 $\pm$ 0.59	-5.24 $\pm$ 0.59	70.83 $\pm$ 0.67	-9.11 $\pm$ 0.90	71.44 $\pm$ 1.61	-8.54 $\pm$ 1.71	74.30 $\pm$ 0.88	0.52 $\pm$ 6.27
EWC	67.36 $\pm$ 0.91	-7.67 $\pm$ 0.86	67.38 $\pm$ 1.23	-11.94 $\pm$ 1.44	63.60 $\pm$ 1.34	-12.25 $\pm$ 1.51	68.97 $\pm$ 0.49	-7.23 $\pm$ 0.57
GEM	46.57 $\pm$ 2.67	-30.22 $\pm$ 3.11	46.28 $\pm$ 1.06	-35.42 $\pm$ 1.14	50.31 $\pm$ 2.23	-27.23 $\pm$ 2.47	47.40 $\pm$ 2.81	-31.29 $\pm$ 3.13
MAS	74.87 $\pm$ 0.51	-3.27 $\pm$ 0.41	72.83 $\pm$ 0.24	-6.63 $\pm$ 0.31	72.52 $\pm$ 0.22	-6.76 $\pm$ 0.24	74.96 $\pm$ 0.63	-3.34 $\pm$ 0.42
TWP	74.54 $\pm$ 0.26	-4.99 $\pm$ 0.45	69.29 $\pm$ 0.88	-11.17 $\pm$ 0.96	72.16 $\pm$ 0.57	-8.32 $\pm$ 0.84	74.72 $\pm$ 0.30	-5.40 $\pm$ 0.44
ER-GNN	47.63 $\pm$ 4.99	-34.91 $\pm$ 5.75	46.63 $\pm$ 1.32	-36.55 $\pm$ 1.66	52.89 $\pm$ 3.57	-30.01 $\pm$ 4.12	44.74 $\pm$ 4.86	-38.35 $\pm$ 5.64
SSM	48.35 $\pm$ 5.82	-34.14 $\pm$ 6.57	44.83 $\pm$ 5.15	-38.81 $\pm$ 5.83	52.05 $\pm$ 2.68	-30.95 $\pm$ 2.95	45.57 $\pm$ 4.65	-37.69 $\pm$ 5.40
TPP	42.13 $\pm$ 0.64	0.00 $\pm$ 0.00	41.90 $\pm$ 0.57	0.00 $\pm$ 0.00	42.92 $\pm$ 0.84	0.00 $\pm$ 0.00	42.34 $\pm$ 0.58	0.00 $\pm$ 0.00
PDGNNs	45.96 $\pm$ 1.09	-0.11 $\pm$ 1.01	48.17 $\pm$ 0.65	-0.73 $\pm$ 1.19	47.10 $\pm$ 0.90	-8.60 $\pm$ 0.88	46.29 $\pm$ 0.86	-0.64 $\pm$ 1.36
LoRAMoE	40.21 $\pm$ 1.54	-30.97 $\pm$ 1.15	34.31 $\pm$ 2.22	-27.35 $\pm$ 3.03	37.47 $\pm$ 1.54	-26.64 $\pm$ 1.98	36.71 $\pm$ 1.98	-27.26 $\pm$ 2.38
Joint	74.79 $\pm$ 0.31	-3.01 $\pm$ 0.40	74.65 $\pm$ 0.72	-2.97 $\pm$ 0.54	74.69 $\pm$ 0.19	-2.93 $\pm$ 0.35	74.90 $\pm$ 0.26	-1.04 $\pm$ 0.58
G-CORMoL	77.26 $\pm$ 0.32	0.49 $\pm$ 0.18	76.16 $\pm$ 0.12	0.15 $\pm$ 0.11	76.04 $\pm$ 0.27	0.13 $\pm$ 0.22	76.69 $\pm$ 0.28	0.27 $\pm$ 0.10

**Figure 3: Average performance during cross-domain continual fine-tuning process.**

4.2.1 Overall Performance Comparison (RQ1). The primary results for overall performance are presented in Table 1 and Figure 3, which compares the average accuracy of G-CORMoL against all baselines across the full sequence of domains. The results clearly demonstrate the superiority of our proposed framework. As shown in Figure 3, most baselines—ranging from traditional CL methods like EWC to specialized CGL approaches like PDGNNs—suffer a steep decline in accuracy as more domains are encountered. This trend highlights their inability to deal with the significant distribution shifts inherent in our challenging multi-domain setting. In contrast, G-CORMoL maintains stable performance throughout the sequence. Its accuracy curve remains consistently high and nearly flat, showing that the model learns new tasks while retaining performance on earlier ones. Consequently, the performance gap between G-CORMoL and the baselines progressively widens throughout the continual learning sequence. Most notably, G-CORMoL not only significantly outperforms all continual learning baselines but also surpasses the Joint-Training upper bound in multiple orders. This notable result suggests that standard joint training can suffer from performance degradation due to conflicting signals between diverse domains. Our method’s modular and orthogonal architecture is designed to mitigate this exact issue.

4.2.2 Analysis of Forgetting and Knowledge Transfer (RQ2). To provide a more fine-grained analysis of the model’s learning dynamics, we present a heatmap in Figure 4 that visualizes the final accuracy on each task after the full sequence has been learned. In this visualization, darker shades represent higher accuracy, indicating strong knowledge retention and transfer, while lighter shades signify performance degradation due to catastrophic forgetting. The heatmaps for the baseline methods are characterized by a pronounced lightening of color in the cells corresponding to earlier tasks. This visually confirms their struggle with catastrophic forgetting, as performance on initial domains degrades significantly after training on later, distribution-shifted tasks. Conversely, the heatmap for G-CORMoL displays a consistently dark coloration across the entire matrix. This provides two critical insights: First, the dark diagonal confirms the complete mitigation of catastrophic forgetting, a direct consequence of our Orthogonal Knowledge Isolation mechanism which structurally prevents inter-task interference. Second, and more importantly, the darkening of cells above the main diagonal provides clear evidence of positive knowledge transfer, where learning subsequent tasks enhances performance on earlier ones. This empirically supports the effectiveness of our Symmetric Dual-Driven Router, demonstrating that it is not merely

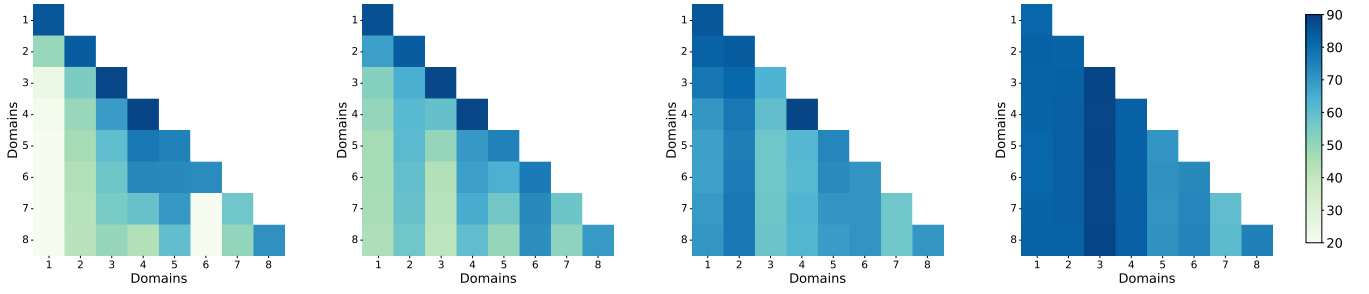


Figure 4: Performance matrices comparing different methods. From left to right: ERGNN, SSM, EWC, and G-CORMoL.

preserving knowledge, but is actively composing synergistic experts to reinforce and improve upon previously acquired skills.

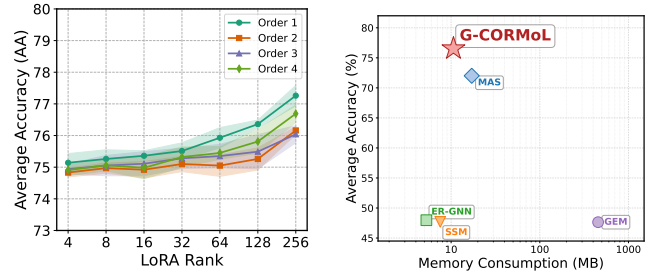
4.2.3 Ablation Studies (RQ3). To isolate the contribution of each key component in G-CORMoL, we conduct a series of ablation studies, with results presented in Table 2. We systematically evaluate the effectiveness of the orthogonal LoRA mechanism and the dual objectives of our router. The results clearly validate our design. First and foremost, the removal of orthogonalization leads to the most pronounced degradation, confirming that guaranteeing knowledge preservation is the essential foundation of our approach. Moreover, orthogonal subspaces also provide the structural basis that makes expert collaboration via the router stable. Next, we find that an untrained router (w/o $\mathcal{L}_{transfer}$ & $\mathcal{L}_{cons}$) also performs poorly, demonstrating the necessity of a learned routing mechanism. When ablating the router’s two objectives individually, we observe that both are beneficial. The performance drop is more substantial when removing the utility prediction loss (w/o $\mathcal{L}_{transfer}$) than when removing the contrastive regularizer (w/o $\mathcal{L}_{cons}$). This suggests that accurately predicting expert utility is the router’s primary function, while the contrastive pressure to specialize acts as a valuable, complementary regularizer.

Table 2: The ablation study of different modules.

Methods	Order1	Order2	Order3	Order4
w/o Orthogonal LoRA	73.71 $\pm$ 1.04	73.91 $\pm$ 1.29	75.62 $\pm$ 0.40	73.04 $\pm$ 1.03
w/o $\mathcal{L}_{transfer}$	76.10 $\pm$ 0.50	75.31 $\pm$ 0.43	75.12 $\pm$ 0.58	75.64 $\pm$ 0.53
w/o $\mathcal{L}_{cons}$	76.59 $\pm$ 0.41	75.24 $\pm$ 0.42	75.21 $\pm$ 0.43	75.73 $\pm$ 0.45
w/o $\mathcal{L}_{transfer}$ & $\mathcal{L}_{cons}$	75.89 $\pm$ 0.62	74.86 $\pm$ 0.72	74.87 $\pm$ 0.68	75.02 $\pm$ 0.41
G-CORMoL	77.26 $\pm$ 0.32	76.16 $\pm$ 0.12	76.04 $\pm$ 0.27	76.69 $\pm$ 0.28

4.2.4 Hyperparameter Experiments (RQ4). We conduct a sensitivity analysis on the LoRA rank r to evaluate its impact on performance. The results, shown in Figure 5(a), reveal a consistent trend: accuracy generally improves across all task orders as the rank increases. A higher rank allows the LoRA matrices to capture more complex and fine-grained information for each task, which naturally leads to better adaptation. Importantly, the performance gains are gradual, indicating that our method is robust and not overly sensitive to this parameter setting. Increasing the rank r yields substantial accuracy improvements while incurring only minimal additional parameter and memory cost, as also illustrated

in Figure 5(b). Moreover, further sensitivity experiments on the loss weights are reported in Appendix C.3.



(a) Performance across different LoRA ranks. (b) Memory consumption comparison with its accuracy.

Figure 5: Comprehensive analysis of hyperparameter sensitivity and memory efficiency.

5 Conclusion

In conclusion, G-CORMoL provides a solution to the critical challenge of cross-domain continual fine-tuning for graph foundation models. The main difficulty lies in handling cross-domain shifts while retaining past knowledge and enabling positive transfer. To tackle this, we develop a framework that enforces orthogonal LoRA subspaces for knowledge preservation and employs a symmetric dual-driven router for dynamic expert composition. Experiments on cross domain data orders show that G-CORMoL effectively alleviates forgetting and promotes positive knowledge transfer, leading to consistent gains over existing methods.

Acknowledgments

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A Appendix

A.1 Proof of Theorem 3.1

The objective of the contrastive regularization is to maximize the conditional Mutual Information (MI) between the expert representation space Z (with instances z) and the graph context space C (with instances $\mathcal{G}_t$), given the router's parameters ϕ . The MI is defined as:

$$I(Z; C | \phi) = \mathbb{E}_{p(z, \mathcal{G}_t | \phi)} \left[\log \frac{p(z | \mathcal{G}_t, \phi)}{p(z | \phi)} \right], \quad (10)$$

Here $p(z | \mathcal{G}_t, \phi)$ denotes the positive distribution, while $p(z | \phi)$ denotes the negative distribution. This is optimized via our contrastive loss $\mathcal{L}_{\text{cons}}$, which trains a discriminator to distinguish one positive sample from $|\mathcal{N}_k|$ negatives. The optimal discriminator for such a task approximates the posterior probability $P(d = 1 | z, \mathcal{G}_t, \phi)$. This posterior is derived using Bayes' rule:

$$\begin{aligned} P(d = 1 | z, \mathcal{G}_t, \phi) &= \frac{p(z | \mathcal{G}_t, \phi)P(d = 1)}{p(z | \mathcal{G}_t, \phi)P(d = 1) + p(z | \phi)P(d = 0)} \\ &= \frac{p(z | \mathcal{G}_t, \phi)}{p(z | \mathcal{G}_t, \phi) + |\mathcal{N}_k|p(z | \phi)}, \end{aligned} \quad (11)$$

To connect this back to MI, we rearrange Equation 11 to isolate the density ratio:

$$\frac{p(z | \mathcal{G}_t, \phi)}{p(z | \phi)} = |\mathcal{N}_k| \cdot \frac{P(d = 1 | z, \mathcal{G}_t, \phi)}{1 - P(d = 1 | z, \mathcal{G}_t, \phi)}, \quad (12)$$

Substituting this identity into the MI definition expands the expression:

$$\begin{aligned} I(Z; C | \phi) &= \mathbb{E}_{p(z, \mathcal{G}_t | \phi)} \left[\log \left(|\mathcal{N}_k| \frac{P(d = 1 | z, \mathcal{G}_t, \phi)}{1 - P(d = 1 | z, \mathcal{G}_t, \phi)} \right) \right] \\ &= \log(|\mathcal{N}_k|) + \mathbb{E}_{p(z, \mathcal{G}_t | \phi)} [\log P(d = 1 | z, \mathcal{G}_t, \phi)] \\ &\quad - \mathbb{E}_{p(z, \mathcal{G}_t | \phi)} [\log(1 - P(d = 1 | z, \mathcal{G}_t, \phi))], \end{aligned} \quad (13)$$

Taking expectation under the standard NCE sampling distribution (one positive and $|\mathcal{N}_k|$ i.i.d. negatives), a key result from NCE theory provides a lower bound for the last term:

$$-\mathbb{E}_{p(z, \mathcal{G}_t | \phi)} [\log(1 - P(d = 1 | z, \mathcal{G}_t, \phi))] \geq -\log \frac{|\mathcal{N}_k| + 1}{|\mathcal{N}_k|}, \quad (14)$$

Substituting this inequality into Eq. 13, we derive the detailed lower bound:

$$\begin{aligned} I(Z; C | \phi) &\geq \log |\mathcal{N}_k| + \mathbb{E}_{p(z, \mathcal{G}_t | \phi)} [\log P(d = 1 | z, \mathcal{G}_t, \phi)] - \log \frac{|\mathcal{N}_k| + 1}{|\mathcal{N}_k|} \\ &= \log |\mathcal{N}_k| + \mathbb{E}_{p(z, \mathcal{G}_t | \phi)} [\log P(d = 1 | z, \mathcal{G}_t, \phi)] \\ &\quad - (\log(|\mathcal{N}_k| + 1) - \log |\mathcal{N}_k|) \\ &= 2 \log |\mathcal{N}_k| - \log(|\mathcal{N}_k| + 1) \\ &\quad + \mathbb{E}_{p(z, \mathcal{G}_t | \phi)} \left[\log \frac{p(z | \mathcal{G}_t, \phi)}{p(z | \mathcal{G}_t, \phi) + |\mathcal{N}_k|p(z | \phi)} \right], \end{aligned} \quad (15)$$

Noting that $2 \log |\mathcal{N}_k| - \log(|\mathcal{N}_k| + 1) = \log |\mathcal{N}_k| - \log(|\mathcal{N}_k| + 1) + \log |\mathcal{N}_k|$, which differs only by an additive $\log |\mathcal{N}_k|$ constant, Eq. 15 can be equivalently rewritten as

$$I(Z; C | \phi) \geq \mathbb{E} \left[\log \frac{p(z | \mathcal{G}_t, \phi)}{p(z | \mathcal{G}_t, \phi) + |\mathcal{N}_k|p(z | \phi)} \right] - \log(|\mathcal{N}_k| + 1), \quad (16)$$

which matches Theorem 3.1. $\square$

This derivation shows that minimizing $\mathcal{L}_{\text{cons}}$ is equivalent to maximizing a variational lower bound on the mutual information. It drives functional specialization: experts co-activated for the same context are pulled closer, while irrelevant ones are pushed apart, yielding a clearer and less redundant expert library. This organization stabilizes the router's behavior and makes expert selection more reliable. On this basis, the utility is learned through supervision, while contrastive regularization enables the router and experts to co-evolve toward a more adaptive routing policy.

B Algorithm and Complexity Analysis

B.1 Algorithm Pipeline

Algorithm 1 summarizes the training and inference procedures of G-CORMoL, covering both the orthogonal expert adaptation process and the symmetric dual-driven routing mechanism.

B.2 Time Complexity Analysis

We analyze the time complexity of G-CORME compared to PDGNN, a state-of-the-art baseline for continual graph learning. Let N_t denote the number of nodes in task $\mathcal{D}_t$, d be the average node degree, L be the number of GNN layers, and r be the LoRA rank. For G-CORME, the orthogonal subspace construction via SVD requires $\mathcal{O}(N_t d^2)$ operations once per task. During training, updating the LoRA matrices $B_t^{(l)}$ has complexity $\mathcal{O}(N_t \cdot r \cdot d)$ per epoch where $r \ll d$. The router optimization adds $\mathcal{O}(t^2)$ overhead for t experts. Overall, the time complexity of G-CORME is $\mathcal{O}(N_t \cdot r \cdot d)$, scaling linearly with the rank r . In contrast, PDGNN requires storing full computation ego-subnetworks with $\mathcal{O}(N_t d^L)$ space complexity. During training, aggregating features across these exponentially-sized neighborhoods results in $\mathcal{O}(N_t d^L \cdot d)$ time complexity per epoch. The d^L factor represents the exponentially growing L -hop neighborhood, which becomes prohibitive for deep GNNs on dense networks. By eliminating this exponential dependency and replacing it with linear scaling in r , G-CORME achieves significantly lower computational complexity than PDGNN.

C Additional Experiments

C.1 G-CORMoL's Knowledge Transfer Analysis

To analyze how G-CORMoL facilitates knowledge transfer, we provide three key visualizations. Figure 7 presents a detailed heatmap of the router's expert selections in the final inference stage. The emergent collaboration patterns are then summarized in the enhancement relation shown in Figure 6. Finally, Figure 8 isolates the incremental gain from each learning step, showing the total accuracy improvement on prior tasks when advancing from stage $t - 1$ to t .

Algorithm 1: The G-CORMoL Framework

```

// - Training Phase -
for each new task  $t = 1, \dots, T$  do
  Initialize orthogonal expert  $\Delta\Theta_t$ ;
  for each training step do
    Sample batch  $(\mathcal{G}_t, y_t)$  from task data  $\mathcal{D}_t$ ;
    // 1. Graph LoRA Adaptation
     $\mathcal{L}_{\text{task}} \leftarrow \text{Loss}(f(\mathcal{G}_t; \Delta\Theta_t), y_t)$ ;
    // 2. Symmetric Dual-Driven Optimization
     $\mathbf{h}_{\mathcal{G}_t} \leftarrow \text{POOL}(f(\mathcal{G}_t))$ ;
    for expert  $k = 1, \dots, t$  do
       $\{g_{k \rightarrow j}\} \leftarrow \mathcal{R}_\phi(\mathbf{h}_{\mathcal{G}_t}, k)$ ;
      Compute  $\mathcal{L}_{\text{transfer}}^{(k)}$ ;
      Compute  $\mathcal{L}_{\text{cons}}^{(k)}$ ;
    end
    // 3. Joint Optimization
     $\mathcal{L}_{\text{total}} \leftarrow$ 
       $\mathcal{L}_{\text{task}} + \lambda_{\text{transfer}} \frac{1}{t} \sum_{k=1}^t \mathcal{L}_{\text{transfer}}^{(k)} + \lambda_{\text{cons}} \frac{1}{t} \sum_{k=1}^t \mathcal{L}_{\text{cons}}^{(k)}$ ;
    Update parameters of  $\theta_{B_t}$ ,  $\phi$  and  $\psi_t$  using  $\nabla \mathcal{L}_{\text{total}}$ ;
  end
end

// - Inference Phase -
Input: Graph  $\mathcal{G}_{in}$  for task  $k$ .
Output: Prediction  $y_{\text{pred}}$ .
 $\mathbf{h}_{\mathcal{G}_{in}} \leftarrow \text{POOL}(f(\mathcal{G}_{in}))$ ;
 $\{g_{k \rightarrow j}\} \leftarrow \mathcal{R}_\phi(\mathbf{h}_{\mathcal{G}_{in}}, k)$ ; // Collaboration scores
 $\mathcal{A} \leftarrow \{j \mid g_{k \rightarrow j} > \tau, j \neq k\}$ ; // Select auxiliary experts
 $\{w_j\}_{j \in \mathcal{A}} \leftarrow \text{Softmax}(\{g_{k \rightarrow j}\}_{j \in \mathcal{A}})$ ;
 $\Delta\Theta_{\text{aux}} \leftarrow \sum_{j \in \mathcal{A}} w_j \cdot \Delta\Theta_j$ ;
 $\Delta\Theta_{\text{total}} \leftarrow \Delta\Theta_k + \Delta\Theta_{\text{aux}}$ ;
 $y_{\text{pred}} \leftarrow f(\mathcal{G}_{in}; \Delta\Theta_{\text{total}})$ ;
return  $y_{\text{pred}}$ ;

```

C.2 LoRA Expert Activation Threshold

We studied the effect of the expert activation threshold, τ , which controls how many auxiliary experts are selected during inference. Figure 9 shows that the model’s performance is stable for a range of threshold values. The accuracy gently increases as τ rises to nearly 0.4, and then gradually declines as the threshold becomes overly restrictive. This suggests that our method is robust to the choice of τ . We set $\tau = 0.4$ in all experiments.

C.3 Loss Weights Analysis

Our framework is relatively robust to the choice of the router’s loss weights, $\lambda_{\text{transfer}}$ and λ_{con} . As shown in Figure 10, performance remains stable across a wide range of values for both hyperparameters, indicating that G-CORMoL does not require extensive tuning. For our main experiments, we set $\lambda_{\text{transfer}} = 0.5$ and $\lambda_{\text{con}} = 0.1$.

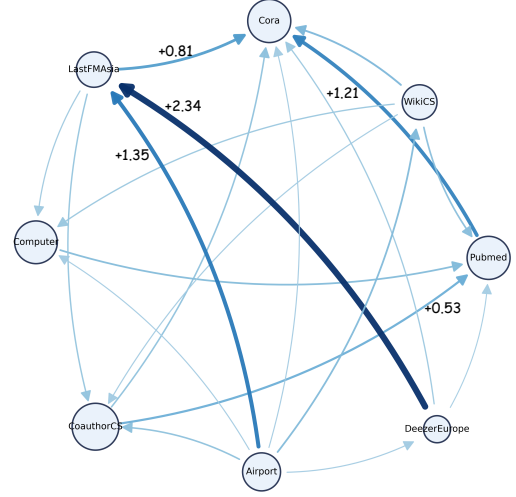
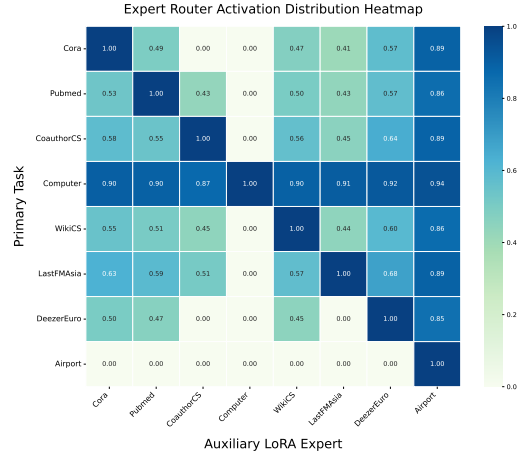
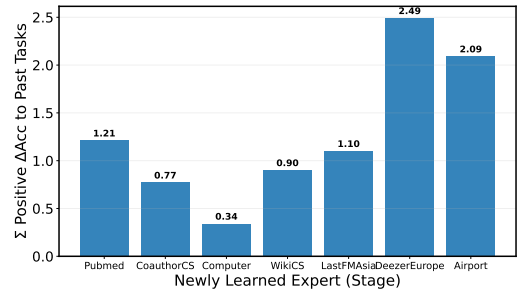
**Figure 6: Knowledge Transfer Flow between Tasks.****Figure 7: Expert Activation Heatmap.****Figure 8: Overall Information Gains.****C.4 Backbone Analysis**

Figure 11 presents results across different GNN backbones in the cross-domain continual fine-tuning setting. Our method maintains consistently high performance on all backbones, while bare fine-tuning suffers drastic performance drops due to forgetting.

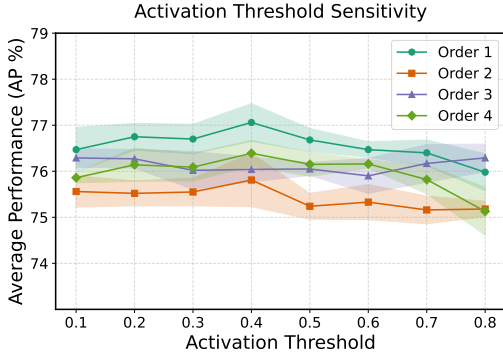


Figure 9: LoRA Experts Activation Threshold Analysis.

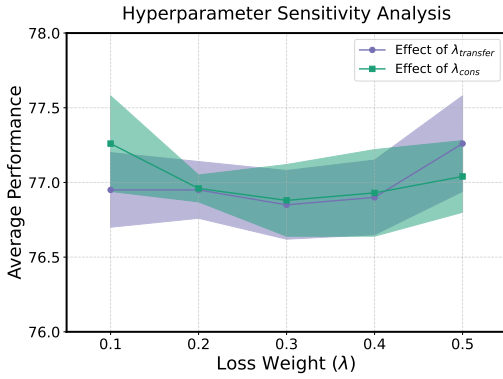


Figure 10: Loss Weights Analysis.

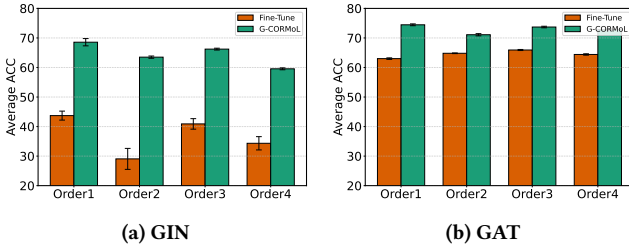


Figure 11: Performance in different GNN backbones.

D Dataset details

D.1 Dataset Statistics

To maintain a consistent experimental setup across these diverse domains, all node features are uniformly projected to a 512-dimensional space, and each dataset is partitioned into training, validation, and test sets following a 60%/20%/20% split. The dataset statistics are reported in Table 3.

D.2 Multi-domain Dataset Orders

To comprehensively evaluate the model’s continual learning capabilities, we construct four task sequences, each designed to probe a different challenge:

- **Order 1:** Cora → Pubmed → CoauthorCS → Computer → WikiCS → LastFMAsia → DeezerEurope → Airport. Provides a standard reference sequence for continual adaptation.

Table 3: Statistics of datasets used in pre-training and continual fine-tuning, grouped into Academic, Co-purchase, Social, and Infrastructure domains.

Domain	Dataset	#Nodes	#Edges	#Feats	#Classes
Academic	Citeseer	3,327	4,732	3,703	6
	DBLP	17,716	105,734	1,639	4
	Cora	2,708	5,429	1,433	7
	Pubmed	19,717	44,338	500	3
	CoauthorCS	18,333	163,788	6,805	15
Co-purchase	WikiCS	11,701	431,726	300	10
	Photo	7,650	119,081	745	8
Social	Computer	13,752	245,778	767	10
	Facebook	22,470	342,004	128	4
	GitHub	37,700	578,006	128	2
	LastFMAsia	7,624	55,612	128	18
	DeezerEurope	28,281	185,504	128	2
Transportation	Airport	3,188	18,631	4	4

- **Order 2:** LastFMAsia → CoauthorCS → Cora → Pubmed → Computer → Airport → WikiCS → DeezerEurope. Designed to test robustness under progressively weaker homophily levels.
- **Order 3:** Airport → DeezerEurope → LastFMAsia → WikiCS → Computer → CoauthorCS → Pubmed → Cora. Evaluates how different knowledge acquisition trajectories affect adaptation.
- **Order 4:** Computer → DeezerEurope → CoauthorCS → WikiCS → LastFMAsia → Pubmed → Cora → Airport. Presents strong inter-task interference, maximizing the potential for forgetting.

E Baseline Details

- **EWC:** Regularization method that applies a quadratic penalty to constrain changes on parameters critical for past tasks.
- **MAS:** Estimates parameter importance via output sensitivity and regularizes updates on the most influential weights.
- **GEM:** Replay-based method that stores a memory buffer of past data and constrains gradients to avoid conflicts with previous tasks.
- **LwF:** Uses knowledge distillation to align current outputs with those of earlier models, preserving knowledge without replay.
- **TWP:** Graph-specific regularization that protects parameters essential for both topology and predictive accuracy on prior graphs.
- **ER-GNN:** Graph replay method that retains representative nodes from past tasks to preserve information.
- **SSM:** Extends node replay by also storing sparsified local subgraphs to capture structural context.
- **TPP:** Prompt-based approach that adapts a frozen GNN via learnable prompts, avoiding modification of backbone parameters.
- **PDGNNs:** Parameter-isolation framework that allocates task-specific modules while keeping shared components fixed.
- **LoRAMoE:** Mixture-of-Experts method that combines multiple LoRA modules through a router; experts specialize to subsets of tasks while the frozen backbone ensures stability.